

CSE_2323

Sequential circuit State equation

Digital Logic Design

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Credit Hour: 3

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Example-1

Question:

Consider the clocked sequential circuit shown in Figure 6-16. This circuit consists of two D flip-flops, A and B, an input 'x', and an output 'y'.

1. Derive the **next-state equations** for flip-flops A and B, and the **output Boolean function** for 'y'.

2. Construct the **state table** for this sequential circuit.

3. Draw the corresponding **state diagram**.

4. Implement the same sequential circuit using **JK flip-flops**.

derive the necessary **flip-flop input functions** (J_A , K_A , J_B , K_B) and Logic diagram

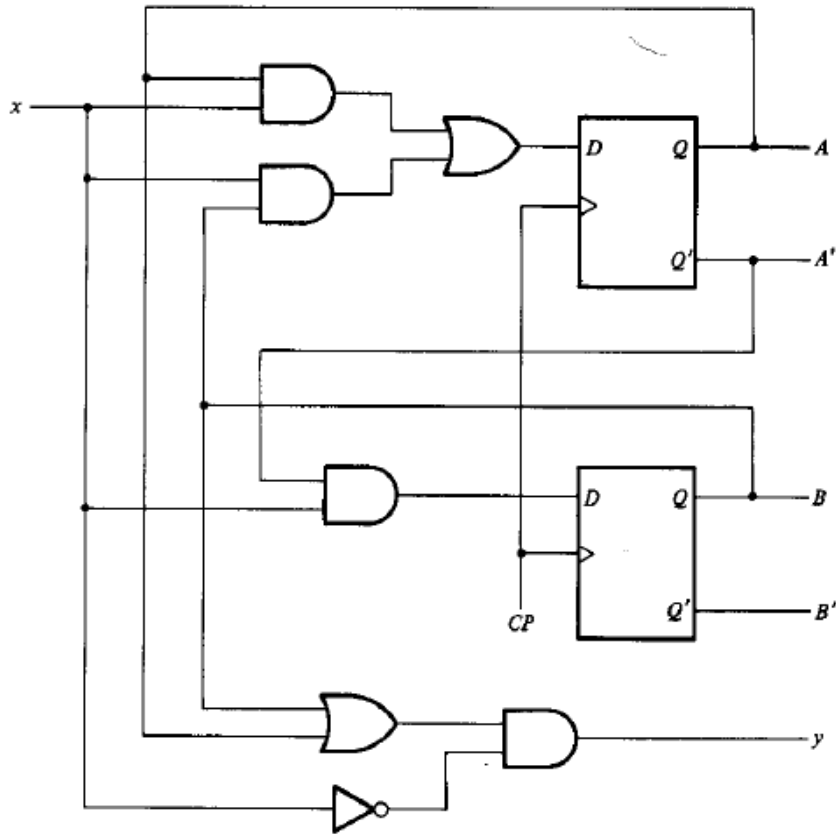


FIGURE 6-16

Example-1

1. Next-State Equations and Output Boolean Function

The D inputs of the flip-flops determine their next states. By analyzing the combinational logic connected to the D inputs of flip-flops A and B in Figure 6-16, we can derive the next-state equations. Similarly, the output 'y' can be expressed algebraically.

•Next-State Equations:

•For flip-flop A:

$$A(t+1) = Ax + Bx$$

•For flip-flop B:

$$B(t+1) = A'x$$

•Output Boolean Function:

$$y = (A + B)x'$$

Example-1

2.State Table(from equation)

The state table enumerates the time sequence of inputs, outputs, and flip-flop states. We will use the format with "Present State," "Input," "Next State," and "Output" sections. The next-state values are determined from the derived state equations, and the output values from the output Boolean function

Present State	Input	Next State	Output
AB	X	$A+B^+$	y
00	0	00	0
00	1	01	0
01	0	00	1
01	1	11	0
10	0	00	1
10	1	10	0
11	0	00	1
11	1	10	0

Example-1

3. State diagram (from state table)

The state diagram graphically represents the information in the state table. Each circle represents a state, identified by the binary number inside (AB). Directed lines show transitions, labeled with "Input/Output".

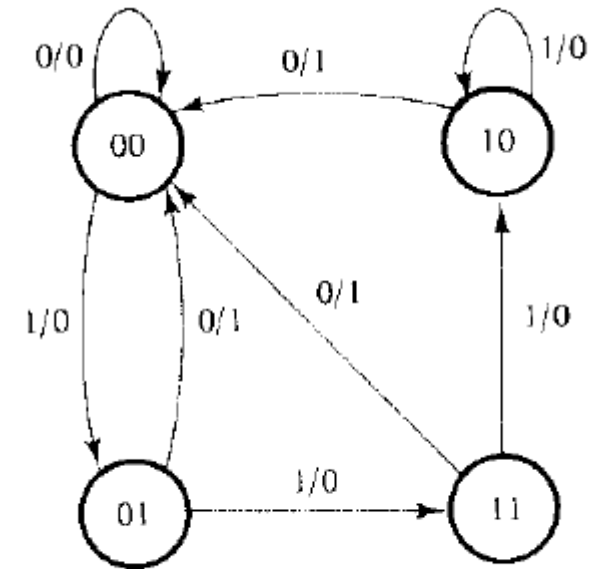


FIGURE 6-17

State diagram of the circuit of Fig. 6-16

Example-1

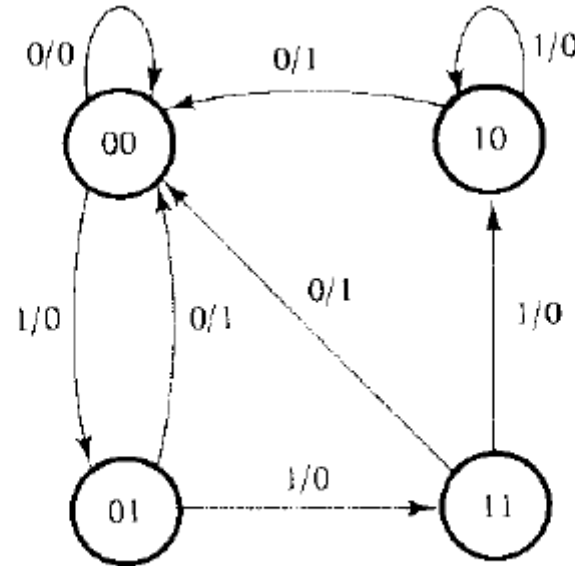


FIGURE 6-17

State diagram of the circuit of Fig. 6-16

4. Let state diagram is given. To implement the same sequential circuit (**defined by the state table/diagram**) using JK flip-flops, we need to derive the flip-flop input functions (J_A , K_A , J_B , K_B) for each flip-flop.

Example-1

We will use the excitation table that combines the present state, input, and desired next state from the original D flip-flop state table with the required J and K inputs to achieve that next state

Present State		Input	Next State		output	Flip-Flop A Inputs	Flip-Flop B Inputs
A	B	X	A ⁺	B ⁺	Y	JA KA	JB KB
0	0	0	0	0	0	0 X	0 X
0	0	1	0	1	0	0 X	1 X
0	1	0	0	0	1	0 X	X 1
0	1	1	1	1	0	1 X	X 0
1	0	0	0	0	1	X 1	0 X
1	0	1	1	0	0	X 0	0 X
1	1	0	0	0	1	X 1	X 1
1	1	1	1	0	0	X 0	X 1

Excitation Table:

Q	Q ⁺	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Example-1

Now, we derive the simplified Boolean expressions for J_A , K_A , J_B , and K_B using Karnaugh maps based on the excitation table which has inputs (A, B, X)

A \ BX				
	00	01	11	10
0			1	
1		X	X	X

$$j_A = BX$$

A \ BX				
	00	01	11	10
0	X	X	X	X
1	1			1

$$K_A = X'$$

A \ BX				
	00	01	11	10
0		1	X	X
1			X	X

$$j_B = A'X$$

A \ BX				
	00	01	11	10
0	X	X		1
1	X	X	1	1

$$K_B = A + X'$$

Example-1

Summary of Flip-Flop Input Functions and Output Function:

$$JA = Bx$$

$$KA = x'$$

$$JB = A'x$$

$$KB = A + X'$$

Same as we can derive for $y = (A + B)x'$

Logic Diagram using JK Flip-Flops: class work

Example-2

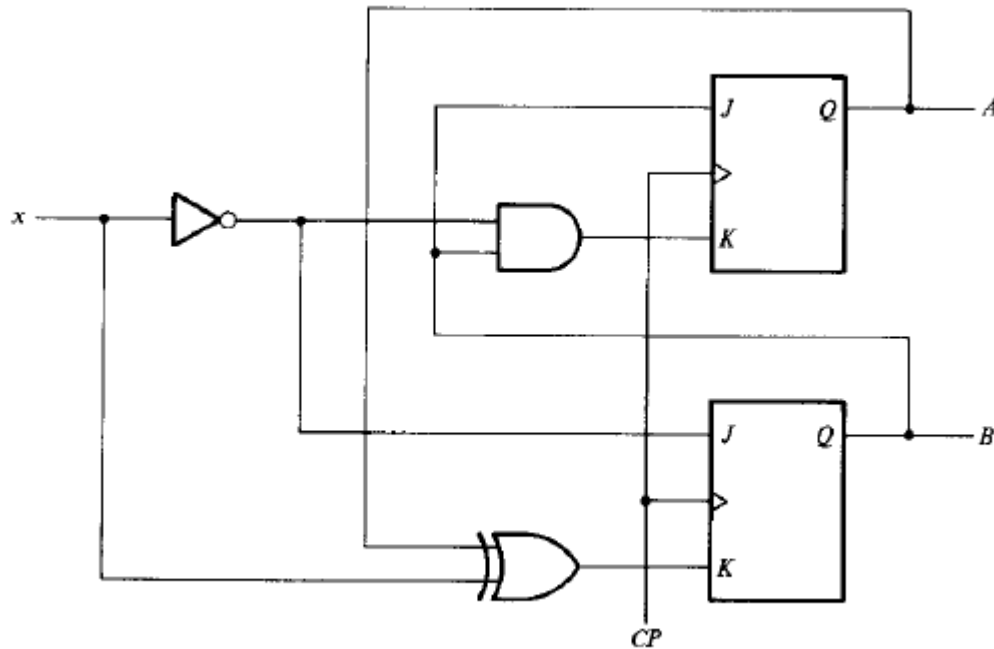


FIGURE 6-19

Consider the sequential circuit with two JK flip-flops A and B and one input x , as shown in Figure 6-19. The circuit has no external outputs.

- Derive the flip-flop input functions for J_A , K_A , J_B , and K_B .
- Construct the state table for this sequential circuit, including the flip-flop input columns and the next-state columns.
- Draw the state diagram for this circuit.

Example-2

Derive the flip-flop input functions for J_A , K_A , J_B , and K_B .

Looking at the logic diagram in Figure 6-19, we can trace the connections to the J and K inputs of flip-flops A and B.

- **For Flip-Flop A:** The J input of A (J_A) is connected directly to the output of flip-flop B.

- Therefore, **$J_A = B$** .

- The K input of A (K_A) is connected to the output of an AND gate. The inputs to this AND gate are the output of flip-flop B and the inverted input x (x'). Therefore, **$K_A = Bx'$** .

- **For Flip-Flop B:**

- The J input of B (J_B) is connected directly to the inverted input x (x'). Therefore, **$J_B = x'$** .

- The K input of B (K_B) is connected to the output of an XOR gate. The inputs to this XOR gate are the output of flip-flop A and the input x. An XOR gate output is 1 if its inputs are different, and 0 if they are the same. This can be expressed as $A'x + Ax'$. Therefore, **$K_B = A'x + Ax'$** .

Example-2

•State table

Present State		Input	Flip-flop inputs		Next State	
A	B	x	$J_A K_A$	$J_B K_B$	$A(t+1)$	$AB(t+1)$
0	0	0	00	10	0	1
0	0	1	00	01	0	0
0	1	0	11	10	1	1
0	1	1	10	01	1	0
1	0	0	00	11	1	1
1	0	1	00	00	1	0
1	1	0	11	11	0	0
1	1	1	10	00	1	1

Truth table

J	K	$Q(t+1)$
0	0	$Q(t)$
0	1	0
1	0	1
1	1	$Q'(t)$

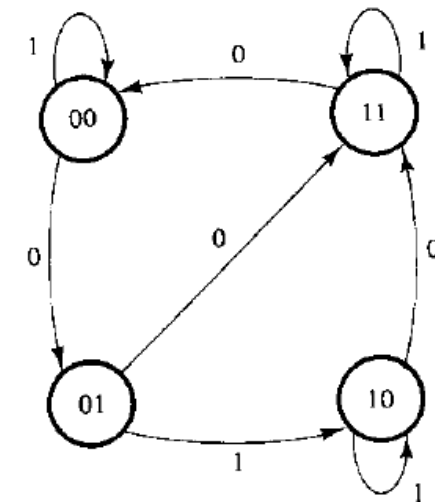


FIGURE 6-20

State diagram of the circuit of Fig. 6-19

Example-3

Question: Design the sequential circuit described by the following state equations using JK flip-flops.

$$A(t+1) = xAB + yA'C + xy$$

$$B(t+1) = xAC + yB'C'$$

$$C(t+1) = x'B + yAB'$$

class work

To design the sequential circuit using JK flip-flops, we need to derive the JK input equations for each flip-flop based on the given state equations. The characteristic equation for a JK flip-flop is $Q(t+1) = JQ'(t) + K'Q(t)$. We'll then use this, along with the provided state equations, to determine the J and K inputs for flip-flops A, B, and C.